

Memory Capacity as a Retrieval Limit: Direct Synaptic Measurement Across a Capacity Range in a Sparse Binary Assembly Network

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ABSTRACT Storage capacity in associative memory is normally treated as a property of the synaptic substrate: patterns superpose until interference makes them unrecoverable. This paper reports a sparse binary assembly network in which that account does not hold over the range where recall fails, established by measuring the synapses directly rather than inferring their state from behaviour. The network has 131,072 neurons and 1.07×10^9 binary synapses with bounded 4-bit counters, contrastive learning and an exact k -winners-take-all readout, and runs on a single consumer GPU. Over the range 24 to 60 stored patterns, at five random seeds per load, the sustained-recall ratio falls by a factor of 3.7 while the functional density of recurrent synapses within stored assemblies remains constant at 80.7–82.2%. The constancy holds separately for neurons exclusive to one assembly (82.7% $\rightarrow$ 81.2%) and for neurons shared across several (69.3% $\rightarrow$ 82.1%, rising) while the shared fraction of each assembly grows from 4.8% to 64.4%, so it is not an artifact of averaging over a degrading sub-population. Retrieval initiation is likewise load-invariant in two independent measurements, and at intermediate load recurrent drive is maintained at unity gain while the retrieved population nonetheless drifts away from the stored assembly. The failure mode is identity drift into a structured neighbourhood: neurons visited during free running but outside the target assembly receive within-assembly recurrent input at $6.9\times$ the population baseline at low load, falling to $1.8\times$ at high load as the composition of that neighbourhood shifts from unallocated neurons to neurons belonging to competing assemblies. Because recall depends on how long retrieval is allowed to run, capacity is reported as a pair: sustained capacity ≈ 36 patterns and transient capacity ≈ 200 .

INDEX TERMS Assembly codes, associative memory, bounded synapses, retrieval dynamics, sparse binary networks, storage capacity.

I. INTRODUCTION

Storage capacity in associative memory is classically an interference problem. In Hopfield and Willshaw-type models, each stored pattern contributes to a shared synaptic matrix, and the cross-talk between patterns grows until retrieval fails [5]. In models with bounded synapses the same logic is sharper still: because synaptic efficacy cannot grow without limit, storing new memories necessarily degrades old ones, and memory lifetime becomes the central quantity [1]. In both traditions the substrate is where capacity lives.

Recent theoretical work has questioned whether this is the whole story. Tamamori’s analysis of kernel Hopfield networks concludes that the practical storage limit is governed not by a lack of geometric separability in feature space but by the loss of dynamical stability against cross-talk noise [3, 4], and Clark [2] shows explicitly that transient recall and persistent (fixed-point) recall are distinct capacities, with patterns retrievable above capacity despite the absence of stable attractors.

Those analyses proceed from dynamics and theory. They do not open the synaptic matrix and inspect it while behaviour fails. The system studied here permits exactly that: the synapses are bounded 4-bit counters whose functional state is a single bit, stored assemblies are explicit sets of neurons, and both can be enumerated exactly at any point during or after learning. We can therefore ask a question that is usually inaccessible: *when recall degrades with load, has the substrate that supports it degraded?*

The answer in this system is no, over the range in which the behavioural transition occurs. Section III presents that measurement and its controls, Section IV the alternative explanations we tested and excluded, and Section V what the result does and does not support.

II. SYSTEM UNDER STUDY

Full implementation detail is in Section VI. This section gives only what is needed to read the results.

A. Population and connectivity

The network comprises 4096 columns of 32 neurons, $N = 131,072$ neurons in total. Columns 0–511 (16,384 neurons) form a clamped input area; the remaining 3584 columns (114,688 neurons) form the recurrent pool in which assemblies are stored. Each neuron has 32 dendritic branches, each sampling a fixed group of 256 presynaptic neurons, giving a fan-in of 8192 and 1.07×10^9 synapses in total. Four branches sample the input area (the afferent path); 28 sample the recurrent pool.

B. Synapses and dynamics

Each synapse is a bounded 4-bit counter stored as four bit-planes. The *functional* weight seen by propagation is the most significant bit: a synapse either transmits or does not. Learning updates the counter; only threshold crossings are visible to the dynamics. Under the contrastive rule the counters are clipped to a symmetric band about the threshold. Total weight memory is 512 MB.

At each timestep each neuron accumulates $\text{popcount}(w \wedge s)$ over its branches, where w is its functional weight vector and s the presynaptic spike pattern. A global threshold selects the 64 columns with the highest drive; within each selected column an exact k -winners-take-all with $k = 24$ selects the firing neurons. This selection rule and the assembly-formation scheme it supports are closest in spirit to the assembly calculus of Papadimitriou et al. [6], which combines a k -cap operation with Hebbian plasticity on random connectivity. The active set is therefore pinned at $64 \times 24 = 1536$ neurons by construction — a fact that matters when interpreting turnover (Section III-F).

C. Learning

Learning is contrastive: with the cue clamped, co-active pairs are potentiated; with the cue released, co-active pairs are depressed. An intrinsic excitability term suppresses recently active neurons during learning and is disabled at readout. Unless stated otherwise all results use the configuration given in Section VI, trained for 40 epochs.

D. Metrics

Overlap between neuron sets A and B is $|A \cap B| / \sqrt{|A||B|}$. *Recall* is the overlap between the retrieved state and the assembly stored for that cue, measured at a 75% cue. *Best-other* is the maximum overlap with any non-target assembly. *Margin* is recall minus best-other. *Sustained ratio* is the overlap with the stored assembly at the last free step divided by that at the first free step, over a 30-step free run; the sustained criterion is a ratio of at least 0.5. Every figure reported below is accompanied by an untrained control run under identical settings.

The sustained ratio is a quotient and becomes uninterpretable when its denominator collapses, so it is reported only under three conditions: (a) mean free-phase recurrent drive to assembly members exceeds twice the untrained floor of 29, (b) recall exceeds best-other, and (c) the matched untrained control does not itself satisfy the sustained criterion. Each condition excludes a distinct degenerate case: (a) collapse of network activity, (b) stable retrieval of a state other than the target, and (c) mechanisms that hold any state regardless of what is stored.

III. RESULTS

A. Recurrent synaptic density is constant across the behavioural transition

We measure *within-assembly functional density*: among all synapses whose presynaptic and postsynaptic neurons both belong to a given stored assembly, the fraction whose counter is above threshold. This is computed exactly, over the 28 recurrent branches only, for every assembly.

Table 1 gives the full capacity sweep and Table 2 the five-seed replication over the range in which the behavioural transition occurs.

Across patterns 24 to 60 the sustained ratio falls by a factor of 3.7 while within-assembly density stays within the band 80.7–82.2%, with per-load standard deviations no larger than 1.12 percentage points. The corresponding five-seed sustained ratio at 12 patterns is 0.672. Density does eventually erode — 78.40% at 74 patterns, 74.27% at 90, 64.94% at 140 — but that erosion begins well past the range in which the behavioural transition occurs, and is therefore a consequence of continued loading rather than its cause.

B. Capacity is reported as a pair

Recall depends on how long free running is allowed to continue. **Sustained capacity** — the highest load at which the mean trace still retains at least half its initial free-run overlap after 30 free steps — is ≈ 36 patterns; the next load tested, 42, does not satisfy the criterion (0.422 ± 0.066). It is a soft crossing rather than a boundary: the ± 1 SD intervals of adjacent loads overlap throughout (36: [0.427, 0.619]; 42: [0.356, 0.488]; 48: [0.243, 0.409]), so no single run identifies which side of the criterion a network falls on. **Transient capacity** — the highest load at which margin at a 75% cue remains positive and the criterion is met — is ≈ 200 patterns. The gap between these two numbers is a property of the measurement horizon, not a discrepancy.

C. The constancy is not an averaging artifact

A mean can conceal a degrading sub-population. The natural candidate is neurons belonging to more than one assembly, which are the most exposed to competing plasticity. We therefore split each assembly’s members into *shared* (belonging to at least one other assembly) and *exclusive*, and measured density separately (Table 3).

The shared fraction of each assembly rises from 4.8% to 64.4% across the load range. Exclusive-member density is flat ($82.69 \rightarrow 81.24\%$) and shared-member density rises ($69.26 \rightarrow 82.09\%$). At low load the few shared neurons are atypical stragglers; by high load sharing is the norm and their density matches the rest. There is no degrading sub-population hidden inside the mean.

D. Retrieval initiation is load-invariant

Two independent quantities measured at the moment of cue release do not vary with load.

First, the overlap lost when the cue is removed. Writing o_5 for the overlap at the last clamp-influenced step and o_6 for the first genuinely free step, the ratio o_6/o_5 is 0.706 ± 0.002 at 12 patterns, 0.728 ± 0.012 at 36, and 0.694 ± 0.022 at 60. Releasing the cue costs a fixed $\approx 29\%$ of assembly overlap regardless of how much is stored.

Second, the recurrent drive received by assembly members at the first free step is 116–120 over patterns 12 to 60 (Table 1); restricted to the exclusive-member subgroup the same quantity is 119–126. We scope this claim to the range of the behavioural transition: beyond it the initial drive does decline (93 at 140 patterns).

Over that range, retrieval initiation is load-blind; load acts only on what follows.

E. At intermediate load, drive is maintained while the trace decays

We define *loop gain* as the ratio of recurrent drive to assembly members between steps of the same parity, two steps apart. Same-parity comparison is required because the free-running dynamics contain a period-2 component (Section III-H); a step-to-step ratio would conflate the oscillation with the trend.

The row for 36 patterns is the important one. Loop gain is 0.977–1.016 — unity within noise — across the entire free run, and the active population is pinned at 1536 neurons by the readout. Yet overlap with the stored assembly decays from 0.720 to 0.520 over the same interval. At intermediate load the

Table 1. Capacity sweep, one run per load. “within” is within-assembly functional density; “drive” is mean recurrent drive to assembly members at the first free step; “criterion” records whether the sustained ratio satisfies the three conditions of Section II. Untrained controls were run at every load and are flat: within-assembly density 6.3613% at all loads, margin -0.029 to -0.079 , drive 28, none satisfying the criterion. Single-run sustained ratios should be read against the replicated values in Table 2.

patterns	within (%)	recall	best-other	margin	sustained	drive	criterion
12	82.09	0.746	0.042	+0.704	0.661	118	met
24	82.41	0.803	0.089	+0.714	0.597	119	met
36	81.34	0.802	0.129	+0.673	0.652	120	met
48	82.07	0.705	0.167	+0.537	0.284	118	met
60	81.78	0.674	0.182	+0.492	0.207	116	met
74	78.40	0.551	0.195	+0.356	0.090	111	met
90	74.27	0.421	0.182	+0.239	0.046	104	met
140	64.94	0.347	0.178	+0.169	0.027	93	met
200	63.43	0.237	0.186	+0.051	0.020	96	met
300	64.52	0.137	0.179	-0.041	0.024	102	<i>not met</i>

Table 2. Five-seed replication. Seeds vary the random weight initialisation, the connectivity and the stimulus set jointly. Density is highly reproducible; the sustained ratio is not. CV is the coefficient of variation of the sustained ratio.

patterns	sustained ratio (mean $\pm$ SD)	margin	within-assembly density (%)	sustained CV
24	0.664 ± 0.051	$+0.755 \pm 0.026$	82.23 ± 0.17	7.7%
36	0.523 ± 0.096	$+0.681 \pm 0.012$	81.42 ± 0.42	18.4%
42	0.422 ± 0.066	$+0.620 \pm 0.026$	80.69 ± 0.63	15.6%
48	0.326 ± 0.083	$+0.577 \pm 0.033$	81.15 ± 0.73	25.5%
60	0.178 ± 0.111	$+0.499 \pm 0.007$	81.49 ± 1.12	62.4%

Table 3. Within-assembly density split by whether the postsynaptic member is shared with another assembly. Both groups hold; the shared group improves.

patterns	shared fraction	density, shared (%)	density, exclusive (%)	difference
12	0.048	69.26	82.69	-13.4
24	0.172	78.46	83.16	-4.7
36	0.366	80.39	81.82	-1.4
48	0.544	82.78	81.31	$+1.5$
60	0.644	82.09	81.24	$+0.8$
48 (untrained)	0.496	6.38	6.35	$+0.03$

Table 4. Loop gain per two-step period across the free run, five seeds. Gain relaxes towards unity at every load, approaching from above at low load and from below at high load. Because gain is not constant across the run, an early-window estimate cannot be extrapolated over the full 30 steps.

patterns	$t_{6 \rightarrow 8}$	$t_{10 \rightarrow 12}$	$t_{14 \rightarrow 16}$	$t_{18 \rightarrow 20}$	$t_{22 \rightarrow 24}$	$t_{26 \rightarrow 28}$
12	1.074	1.035	1.014	0.988	1.006	0.986
36	0.998	1.016	0.996	1.008	0.977	0.998
60	0.892	0.950	0.910	0.940	0.965	0.971

trace degrades without any loss of recurrent drive. Whatever is failing, it is not amplitude.

F. The failure mode is identity drift into a structured neighbourhood

Since population size is fixed by the readout and drive is maintained, we measured which neurons are active and how that set changes.

Turnover and leak. Per free step, the fraction of the active set replaced is 31.3% at 12 patterns, 32.9% at 36, and 51.6% at 60; the untrained control replaces 88–96% per step. Over the same interval the overlap with the stored assembly is lost at 0.149 ± 0.062 , 1.199 ± 0.639 and 5.162 ± 1.844 percent per step. The ratio of churn to leak is 255 ± 148 , 37 ± 24 and 11 ± 4 ; these are reported as an ordering rather than as point values, the low-load figure being poorly resolved. What is well resolved is that turnover is essentially identical at 12 and

36 patterns (31.3% versus 32.9%) while leak differs eightfold. Load changes the directedness of churn, not its magnitude.

The active set is not a fixed population. The union of all neurons active at any point during the 30 free steps is 3265 ± 60 at 12 patterns, 4701 ± 702 at 36, and $11,257 \pm 1410$ at 60, against an active set of 1536 — ratios of 2.13, 3.06 and 7.33. The untrained control visits $30,199 \pm 1611$ ($19.66\times$). The trace therefore cycles through a population substantially larger than the readout displays at any instant, and the learned constraint bounds the size of that population — a ninefold tightening relative to untrained — rather than fixing its membership. The union co-varies monotonically with load, as does the sustained ratio; we report it as a descriptive quantity and do not treat the correlation between them as evidence of mechanism, since any two monotone functions of load must correlate.

Table 5. Characterisation of the population visited during free running. “dens. out” is the functional density of recurrent synapses from stored assembly members onto visited-but-non-member neurons; “dens. all” is the same quantity for the whole recurrent pool. “mult.” is the mean number of assemblies a neuron belongs to, given for the outside group and for the whole pool.

	in-assy.	outside	% outside	dens. in (%)	dens. out (%)	dens. all (%)	mult. out / pool
12 patterns	1531	1733	53.1	82.13	59.68	8.62	0.12/0.16
36 patterns	1536	3166	67.3	81.43	25.61	8.74	1.00/0.48
60 patterns	1536	9721	86.4	81.49	16.70	9.10	1.53/0.80
36, untrained	613	29,585	98.0	6.38	6.28	6.25	0.56/0.48

The visited population is structured. Table 5 characterises it. Neurons visited during free running but outside the stored assembly receive within-assembly recurrent input at 59.68% at 12 patterns, against a population baseline of 8.62%: an enrichment of $6.9\times$. In the untrained control the same two numbers are 6.28% and 6.25%, an enrichment of $1.00\times$, confirming that the measurement is not an artifact of the procedure.

Two measurements point the same way from opposite directions. The enrichment of within-assembly input onto visited non-members falls $6.9\times \rightarrow 2.9\times \rightarrow 1.8\times$ with load. Simultaneously, the assembly-multiplicity of those neurons inverts: at 12 patterns they are *less* likely than average to belong to another assembly (0.12 versus a population mean of 0.16), that is, they are mostly unallocated neurons that happen to be well wired to this assembly; by 60 patterns they are nearly twice as likely (1.53 versus 0.80), that is, they belong to competing assemblies.

Load does not merely widen the population the trace explores; it changes that population’s composition from assembly-adjacent to inter-assembly.

G. Allocation

Assemblies are allocated with very low mutual overlap. At 12 patterns the union of all assigned neurons is 18,015 against a sum of assembly sizes of 18,456 (excess 441, i.e. 2.4% reuse), and the mean pairwise assembly Jaccard index is 0.00222 against a chance value of 0.00674 for independently drawn sets of the same size — $3.0\times$ below chance. The untrained control gives 0.00946, which is $1.4\times$ above chance.

The stricter test accounts for forced sharing. At high load, storing many assemblies of 1536 neurons in a pool that the allocator uses only partially requires reuse, so the appropriate null is m^2/U , where m is assembly size and U the realised union. At 140 patterns the measured mean pairwise Jaccard index is 0.00647 against a sharing-adjusted null of 0.00820 ($0.79\times$); at 300 patterns it is 0.00777 against 0.00747 ($1.04\times$), at a mean neuron multiplicity of 4.45. Allocation overlap therefore remains at or below what forced sharing requires even in the heavy-reuse regime.

Competitive learning and k -winners-take-all are standard mechanisms for producing decorrelated sparse codes [7, 8], so sub-chance overlap is close to expected; what is reported here is that decorrelation persists into a regime where sharing is geometrically unavoidable.

Residual best-other overlap is a tail statistic. At 140 and 300 patterns, best-other is 0.178 and 0.179, but the *mean* overlap with all non-target assemblies is 0.025 and 0.026 — close to the sharing-adjusted null, and equal to the untrained floor of 0.025. Best-other is therefore driven by a small number of high-overlap pairs rather than by uniform confusion. Those

pairs correlate with shared cue columns (Pearson $r = +0.403$ and $+0.355$ trained, versus $+0.098$ and $+0.097$ untrained), so the network learns a mapping from input geometry into assembly geometry. Retrieval failure, however, does not track that mapping (Section IV).

H. Period-2 alternation

Free-running states overlap more with the state two steps earlier than with the immediately preceding state, in trained networks only: at 12 patterns $J(t, t-2)$ runs from 0.474 to 0.602 across the free run against $J(t, t-1)$ of 0.485 to 0.526, whereas the untrained control gives 0.025–0.034 and 0.021–0.036 — equal within noise. The alternation is therefore a learned dynamical property rather than a sampling artifact.

It is not, however, the mechanism behind the wide visited population. If free running were a clean two-cycle over disjoint sets, each parity stream would visit ≈ 1536 neurons. Measured, each parity stream alone visits $1.90\text{--}1.93\times$ the active set at 12 patterns, and the two parity sets overlap at 0.892 ± 0.008 : they are nearly the same population. Parity overlap declines monotonically with load ($0.892 \rightarrow 0.814 \rightarrow 0.746 \rightarrow 0.672 \rightarrow 0.617$ over patterns 12 to 60; untrained 0.299), so alternation strengthens as the trace degrades. It accompanies the failure rather than causing it.

IV. ALTERNATIVE EXPLANATIONS TESTED AND EXCLUDED

Each of the following is a candidate explanation for the load-dependent decline in recall. Each was tested and excluded. Untrained controls accompany every entry.

Hub capture. Hypothesis: a small set of highly connected neurons wins the competition regardless of cue. Excluded. The functional in-degree distribution has coefficient of variation 0.150 and $p_{99}/p_{50} = 1.25$ — no heavy tail. Neurons winning in at least 11 of 12 free runs have *lower* in-degree (499.2) than neurons that never win (680.8), a ratio of 0.733, inverted relative to the prediction.

Synaptic erosion at the boundary. Hypothesis: contrastive depression driven by other patterns erodes the recurrent scaffold. Excluded over the relevant range: within-assembly density is 82.09, 82.41, 81.34 and 82.07% at 12, 24, 36 and 48 patterns while the five-seed sustained ratio falls from 0.672 to 0.326 (Section III-A). Erosion occurs later and is a consequence, not the cause.

Allocation interference. Excluded: measured pairwise assembly overlap is $0.79\text{--}1.04\times$ the sharing-adjusted null at up to $4.45\times$ forced multiplicity (Section III-G).

Input-space collisions. Hypothesis: cues drawn from a limited input space overlap, so assemblies must overlap. Excluded by a controlled comparison in which pool size and afferent drive were held fixed and only the expected pairwise cue overlap was varied fourfold (expected shared cue columns 2.0 versus 8.0): margin -0.080 versus -0.086 , mean non-target overlap 0.039 versus 0.041. Indistinguishable.

Pool size. Hypothesis: capacity is limited by the number of available neurons. Excluded at fixed code sparsity: over a sixfold pool range (32,768 to 200,704 recurrent neurons, with the number of active columns scaled to hold the active-column fraction constant at 1.79%), recall at 140 patterns is 0.300, 0.345, 0.347 and 0.325 — flat — while within-assembly density is likewise flat at 60.7–65.3%.

Averaging artifact. Excluded by the shared/exclusive split of Section III-C.

Drive magnitude at intermediate load. Excluded: loop gain is 0.977–1.016 across the entire free run at 36 patterns while overlap decays $0.720 \rightarrow 0.520$ (Section III-E).

Readout-side rescue. Two interventions on frozen weights were tested and neither restores sustained activity in a way attributable to the stored memory. Reducing k does not restore it (best sustained ratio 0.334 at $k = 20$, worse below $k = 16$); it lowers total drive rather than sharpening competition, and the relevant competition is between columns while k operates within one. Adding a facilitation term favouring recently active neurons does raise the sustained ratio ($0.284 \rightarrow 0.814$ at 48 patterns), but the matched untrained control also sustains, at 1.000: the mechanism holds any state. Measured as learned contribution (trained margin minus untrained margin) it *reduces* the learned signal at low load, from $+0.733$ to $+0.268$ at 12 patterns, while raising it at high load, from $+0.038$ to $+0.124$ at 300. It is a load-dependent trade rather than a fix.

V. DISCUSSION

The central result is narrow: in this system, over the load range in which sustained recall collapses, the synapses that support recall do not change. The supporting evidence is that the constancy survives disaggregation into the two sub-populations where degradation would be expected to appear first, that two independent measurements of retrieval initiation are also load-invariant, and that at intermediate load recurrent drive is maintained at unity gain while the trace decays anyway.

What replaces substrate degradation as the explanation is identity drift, and the characterisation in Section III-F makes that specific rather than merely negative: the trace visits a population two to seven times the size of the instantaneous active set, that population is strongly enriched for within-assembly input at low load, and load shifts its composition from unallocated assembly-adjacent neurons towards neurons belonging to competing assemblies. The composition shift, supported by two measurements pointing in opposite directions — connectivity enrichment falling, assembly multiplicity rising — is the most specific mechanistic statement the data support.

A. Relation to bounded-synapse theory

The natural comparison is to the bounded-synapse literature. Fusi and Abbott [1] show that with bounded synapses, storing new memories acts as noise on previously stored ones, and that memory lifetime grows with the square of the number of synaptic states only when potentiation and depression are precisely balanced. Our synapses are bounded (4-bit counters, clipped to a symmetric band, seven effective states) and the learning rule applies potentiation and depression in explicitly paired phases. The measured depression-to-potentiation event ratio is 0.56–0.79 under the configuration used throughout, and 0.87–1.36 at the higher depression setting. This places the system in the bounded-synapse regime their analysis addresses; we do not claim it satisfies the precise potentiation–depression balance their quadratic lifetime result requires.

Their analysis concerns *sequential* storage, in which each new memory perturbs those already stored, whereas we train all patterns *interleaved to equilibrium* over 40 epochs. A reader may reasonably object that overwriting is avoided by construction in our protocol, so finding the substrate intact says nothing about their result. We answer in three parts. First, we concede the protocol difference and do not claim a palimpsest result. Second, the comparison we draw is with the prediction that bounded synapses under continued storage degrade as capacity is approached; we measure the substrate across the range where behaviour fails and find it intact, which is a statement about our system that does not require protocol matching. Third, the shared-member result of Section III-C responds directly to the objection: if interleaving merely avoided overwriting by construction, the neurons shared across many assemblies are precisely where it would show, and they are the group whose density *rises* while the shared fraction grows from 4.8% to 64.4%. The substrate is not immune — density does fall to 74.27% and 64.94% at 90 and 140 patterns — but the behavioural failure precedes it.

B. Limitations

This is a measurement study of one architecture on one machine, and one task family (cued recall of randomly generated column patterns). Whether the dissociation between substrate and retrieval holds under sequential storage, other learning rules, or other readouts is untested. The sustained-capacity boundary is soft, with overlapping per-load distributions, and should not be read as a sharp transition. Throughput figures are reported for context only; no result above depends on a timing measurement.

VI. METHODS

A. Implementation

The network is implemented in Objective-C++ and Metal Shading Language and runs on the integrated GPU of an Apple M1 Pro (16-core GPU, 16 GB unified memory). Shaders are compiled at runtime from embedded source, which makes geometry specialisation by string substitution free; this is used to vary population size without recompiling the host program.

The configuration used throughout unless stated: 4096 columns of 32 neurons; 512 input columns; 32 branches per neuron of which 4 are afferent; 256 presynaptic neurons per branch; initial functional synapse density 2^{-4} ; 64 active columns; $k = 24$ winners per column; dendritic threshold 1;

supralinear branch mode; 64 stimulus columns per pattern; contrastive learning with depression applied on alternate free steps; intrinsic excitability increment 73 and decrement 1; presentation length $T = 10$ steps of which the first 5 are clamped; 40 training epochs. Seeds vary weight initialisation, connectivity and the stimulus set jointly. Throughput is 2554–2956 timesteps per second trained and 3795–4930 untrained, reproducible to $\pm 3\%$ on clean single runs.

B. Verification of primitives

Correctness of the numerical primitives was established before any result was collected. The SIMD ballot primitive was verified lane-exact at a SIMD width of 32 against a known bit pattern. The exact k -winners-take-all returns exactly k set bits for $k \in \{1, 3, 5, 8, 16\}$ across all 4096 columns, at 2.5 ns per column. The bit-serial saturating increment, decrement (borrow path) and symmetric clip were verified against a scalar reference over 2,097,152 random values each, for three clip bands, with zero mismatches. A parallelised top- K column selection was verified bit-identical to the serial implementation over 983,040 winner-set words on an untrained network and 491,520 on a trained one before being adopted.

C. Memory system and sparsity geometry

Random gather in which each lane follows its own pointer reaches 3–5% of peak bandwidth, and making the gathered blocks contiguous barely helps. The same random blocks read cooperatively by all 32 lanes of a SIMD group reach 102 GB/s, a $13.6\times$ improvement at identical block size and identical randomness. All synaptic access is therefore SIMD-group-cooperative, in tiles of 32 postsynaptic by 128 presynaptic binary weights.

With neuron count, fan-in and spike count held identical, activity clustered into columns runs at 14,585 timesteps per second against 1,285 for randomly scattered activity — an $11.4\times$ difference arising purely from the geometry of the sparse code, because a presynaptic group of 128 neurons is silent far more often under columnar activity than under scattered activity at the same overall rate. Dense binary propagation reaches 1.36×10^{12} synapse operations per second.

Two alternative execution strategies were measured and rejected. A persistent kernel with a device-wide barrier is not viable: only 8 threadgroups are reliably co-resident regardless of threadgroup size; ordinary device-memory writes are not visible across threadgroups within a dispatch, so all cross-threadgroup traffic would have to be atomic; and a correct barrier, which requires the spin read to be a read-modify-write, costs $1.7\mu\text{s}$ against $2.5\mu\text{s}$ for simply issuing a separate dispatch per timestep. Distributing computation across the CPU’s matrix coprocessor, neural engine and efficiency cores is also ruled out: a host round trip costs $165\mu\text{s}$ against timesteps of 0.35–0.44 ms, so any per-step offload loses.

Per-kernel timing attribution used cumulative ablation in the production encoder. The device exposes one counter set and does not support counter sampling at dispatch boundaries, and a marginal-cost method is invalid here because repeated back-to-back dispatches of the same kernel run from cache.

D. Data and code availability

The complete implementation (a single Objective-C++ source file containing all Metal kernels), the benchmark and sweep

scripts, and the measurement logs are available at <https://github.com/rakib-nyc/axon>, archived alongside this preprint at 10.5281/zenodo.21881612. Every table in this paper is reproducible from a single command line with the configuration and seed recorded in the corresponding script; the repository README gives the command for each table.

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